THE PÓLYA–VINOGRADOV INEQUALITY WITH $\log\log q$

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Abstract. For every primitive character χ modulo $q \geq 3$ and every x, $\sum_{n\leq x}\chi(n) \ll \sqrt{q}\log\log q$, which is optimal. Montgomery and Vaughan proved this under the generalised Riemann hypothesis. We deduce it from the zero-free half-plane $\operatorname{Re} s > 7/8$ and prove the bound for exponential sums with multiplicative coefficients that it needs. We also prove two consequences of the half-plane obtained earlier by others: every prime p has a prime primitive root $\ll_\eta (\tau(p-1)^2\log p)^{8+\eta}$, and $\sum_{n\leq x}\chi(n) \ll_\varepsilon q^\varepsilon x^{7/8+\varepsilon}$ for nonprincipal χ. All proofs are checked in Lean 4.

1. Introduction

The main result of this paper is Theorem 1.7: for every primitive character the Pólya–Vinogradov inequality holds with the optimal factor $\sqrt{q}\log\log q$. Montgomery and Vaughan proved this under the generalised Riemann hypothesis; we deduce it from the zero-free half-plane $\operatorname{Re} s > 7/8$ (Theorem 1.8). Theorems 1.1, 1.4 and 1.5 are consequences of the same half-plane that are classical deductions or were obtained earlier by others, as explained below; we include them because every result of the paper is proved in Lean 4 from Theorem 1.8 alone.

For a prime p let $g(p)$ be the least primitive root modulo p and $g^*(p)$ the least prime primitive root. Vinogradov conjectured that $g(p) \ll_\varepsilon p^\varepsilon$ for every $\varepsilon > 0$. Burgess [3] proved $g(p) \ll_\varepsilon p^{1/4+\varepsilon}$; for the prime version the least prime in a progression [14] gives $g^*(p) \ll p^{5.5}$. Under the generalised Riemann hypothesis Shoup [25] and Martin [16] proved $g^*(p) \ll \omega^4(\log\omega + 1)^4\log^2 p$ with $\omega = \omega(p-1)$. Erdős [8] asked whether every prime p has a prime $q < p$ that is a primitive root modulo p (Problem 985 of [2]).

Let τ denote the divisor function. Throughout, χ is a Dirichlet character and $L(s,\chi)$ its L-function.

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Theorem 1.1. *For every $\eta > 0$ there is C_η such that every prime p has a prime primitive root ℓ with*

$$\ell \le C_\eta\big(\tau(p-1)^2 \log(p+2)\big)^{8+\eta}.$$

Corollary 1.2. *For every $\varepsilon > 0$ there is C_ε such that every prime p has a prime primitive root $\ell \le C_\varepsilon p^\varepsilon$. In particular $g(p) \le g^*(p) \ll_\varepsilon p^\varepsilon$.*

Corollary 1.3. *There is p_0 such that every prime $p \ge p_0$ has a prime $q < p$ that is a primitive root modulo p.*

Theorem 1.1 is not new in substance, and stronger bounds were obtained from the same half-plane before this paper. Sandlund [24] derived $g^*(p) \le C_\delta\{k^2(1+\log k)^2 \log 2p\}^{1/\delta}$, $k = \omega(p-1)$, from a zero-free half-plane $\operatorname{Re} s > 1 - \delta$ by Shoup's sieve; Guo [11, §6.4] proved $g^*(p) \ll k^{16}(1+\log k)^{16}(\log p)^8 \ll (\log p)^{24}$ from Martin's shifted sieve [16, Prop. 3]; and the notes [28] give an explicit bound $(\Lambda_p \log p)^8$ with a sieve factor $\Lambda_p \le 2^{\omega(p-1)}$. Corollaries 1.2 and 1.3 follow from each of these bounds. Our proof expands the indicator of the primitive roots over all divisors of $p-1$, which costs the factor $\tau(p-1)^2$; what is new here is the complete Lean proof.

Vinogradov also conjectured that $\sum_{n \le x} \chi(n) = o(x)$ for nonprincipal characters χ modulo q once $x \ge q^\varepsilon$; see [10]. For prime moduli Burgess [3] proved this for $x \ge q^{1/4+\varepsilon}$, and it follows from the generalised Riemann hypothesis.

Theorem 1.4. *For every $\delta > 0$ there is C_δ such that for every nonprincipal character χ modulo q and every $s = \sigma + it$ with $\sigma \ge 7/8 + \delta$,*

$$|L(s,\chi)| + |L(s,\chi)|^{-1} \le C_\delta\big(q(3+|t|)\big)^{\delta}.$$

Theorem 1.5. *For every $\delta > 0$ there is C_δ such that for every nonprincipal character χ modulo q and every $x \ge 1$,*

$$\Big|\sum_{n \le x} \chi(n)\Big| \le C_\delta\, q^\delta x^{7/8+\delta}.$$

Corollary 1.6. *For every $\eta > 0$ and $\theta > 7/8$ there is C such that $\big|\sum_{n \le x} \chi(n)\big| \le C x^\theta$ for every nonprincipal character χ modulo q and every $x \ge q^\eta$. In particular $\sum_{n \le x} \chi(n) = o(x)$ uniformly for $x \ge q^\eta$.*

The Pólya–Vinogradov inequality [7, Ch. 23] gives $\big|\sum_{n \le x} \chi(n)\big| \ll \sqrt{q}\log q$ for every nonprincipal χ modulo q. Paley [23] showed that $\log\log q$ cannot replace $\log q$ with a factor tending to 0: for infinitely many primitive quadratic characters the maximum exceeds $c\sqrt{q}\log\log q$. Montgomery and Vaughan [17] proved the upper bound $\sqrt{q}\log\log q$ for primitive characters under the generalised Riemann hypothesis, and Granville and

Soundararajan [9] improved the exponent of $\log q$ unconditionally for characters of fixed odd order.

Theorem 1.7. *There is an absolute constant C such that for every primitive character χ modulo $q \geq 3$ and every $x \geq 1$,*

$$\Big|\sum_{n\leq x}\chi(n)\Big| \leq C\sqrt{q}\log\log q.$$

Montgomery and Vaughan use the generalised Riemann hypothesis only for prime sums of the characters $\chi\psi$, with ψ modulo a small r, on the major arcs; Theorem 1.8 below suffices there. On the minor arcs they use their unconditional bound for exponential sums with multiplicative coefficients; a weaker form, Theorem 6.1, suffices, and we prove it in Section 6.

The input is the zero-free half-plane of [19], from which [19] deduces, as in [1, Thm. 6.7], that the least quadratic nonresidue is $\ll_\varepsilon p^\varepsilon$. A primitive root must avoid every subgroup of prime index at once, and the number of characters involved is the source of the factor $\tau(p-1)^2$.

Theorem 1.8 ([19])**.** *If χ is a nonprincipal Dirichlet character, then $L(s,\chi)\neq 0$ for $\operatorname{Re} s > 7/8$.*

Theorem 1.8 is formalised in Lean 4 in [19]. Theorems 1.1, 1.4 and 1.5 and Corollaries 1.2, 1.3 and 1.6 and Theorems 6.1 and 1.7 are formalised here, with Theorem 1.8 as the only external input (Section 8). The constants are effective.

Theorem 1.4 is Littlewood's deduction of a Lindelöf bound from a zero-free half-plane [27, §14.2], and Theorem 1.5 follows from it by a contour shift; the new ingredient is Theorem 1.8. The notes [28, Hunt 125, §5] record this Lindelöf-type consequence of the half-plane and the resulting $o(x)$ for $\log x \geq (\log q)^{1-\varepsilon}$. Among the consequences of Theorem 1.8, [19] lists the least quadratic nonresidue, the prime number theorem in progressions, class numbers and idoneal numbers; Guo [11, §6] adds the exceptional set in the binary Goldbach problem and the bound $q^{7/3+\varepsilon}$ for the least prime in a progression, also in [28, Hunt 123]. We have not found Theorem 1.7 in these sources. All results here hold with $7/8$ replaced by any $\theta \in [1/2, 1)$ such that $L(s,\chi)\neq 0$ for $\operatorname{Re} s > \theta$ and every nonprincipal χ, for instance by the value $0.874957\ldots$ claimed in [15]. The companion papers [20, 21] prove, from a zero-free strip $\operatorname{Re} s > 1 - 10^{-6}$ for Hecke L-functions of cyclotomic fields containing the twelfth roots of unity, that every integer other than -1 and the squares is a primitive root modulo infinitely many primes, and that polynomials over $\mathbb{F}_p$ can be factored in deterministic polynomial time; neither bounds $g(p)$.

Remark 1.9. The constant p_0 of Corollary 1.3 is effective but far beyond direct computation. A computation (Section 8) gives $g^*(p) < p$ for every

prime $3 \le p \le 10^8$, with $\max g^*(p) = 223$ at $p = 73189117$. Erdős's question therefore remains open for the primes in $[10^8, p_0)$.

Remark 1.10. Since $\tau(n) \le 2^{(1+o(1))\log n/\log\log n}$ [13, Thm. 317], Theorem 1.1 gives $g^*(p) \le \exp\bigl(c \log p/\log\log p\bigr)$ for an absolute constant c. This form is not formalised.

Remark 1.11. Carella has claimed $g^*(p) \ll p^{5/\log\log p}$ [4] and $g^*(p) = O((\log p)^{1+\varepsilon})$ [5, 6] without hypotheses. In [4], the estimate (16) bounds $\sum_{n\le x}\chi(n)\Lambda(n)n^{-2}$ by $O(1/x)$; the sum tends to $-L'(2,\chi)/L(2,\chi)$, which equals $-0.17264\ldots$ for the character modulo 3. In the arXiv version of [5], Lemma 7.1 asserts $E(x,y) \ll yp^{-\varepsilon}$ for the sum $E(x,y)$ of (7.2) when no integer of $[x, x+y]$ is a primitive root. Under that hypothesis every inner sum over m equals -1, so $E(x,y) = -\varphi(p-1)\,\#([x,x+y]\cap\mathbb{Z})/p$ exactly, while $\varphi(p-1)/p \gg 1/\log\log p$; the step (7.2) replaces $\sum_n e(m\tau^n/p)$ by its value at $m = 1$. We could not obtain the text of [6].

2. The prime sum of a nonprincipal character

For an arithmetic function f and $x \ge 1$ put

$$S(f;x) = \sum_{n\le x} f(n)\Lambda(n)\Bigl(1 - \frac{n}{x}\Bigr),$$

and write $S(\chi;x)$ when $f = \chi$. The aim of this section is Proposition 2.6.

Lemma 2.1. *Let $a\colon \mathbb{N} \to \mathbb{C}$ satisfy $\sum_n |a(n)|n^{-2} < \infty$ and put $F(s) = \sum_n a(n)n^{-s}$ for $\operatorname{Re} s \ge 2$. For $x > 0$,*

$$\sum_{n\le x} a(n)\Bigl(1 - \frac{n}{x}\Bigr) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(2+it)\,\frac{x^{2+it}}{(2+it)(3+it)}\,dt.$$

Proof. For $y > 0$ let $k(y) = \frac{1}{2\pi i}\int_{(2)} y^s\,\frac{ds}{s(s+1)}$. If $y \ge 1$, moving the line to the left through the poles at 0 and -1 gives $k(y) = 1 - 1/y$; if $y < 1$, moving it to the right gives $k(y) = 0$. Since $|s(s+1)|^{-1}$ is integrable on $\operatorname{Re} s = 2$ and $\sum_n |a(n)|n^{-2} < \infty$, summation and integration may be exchanged: the right side equals $\sum_n a(n)k(x/n)$. □

Lemma 2.2. *Let χ be nonprincipal modulo q. For $\operatorname{Re} s = \sigma > 0$, $|L(s,\chi)| \le q|s|/\sigma$.*

Proof. Put $A(t) = \sum_{n\le t}\chi(n)$; then $|A(t)| \le q$, and partial summation gives $L(s,\chi) = s\int_1^\infty A(t)t^{-s-1}\,dt$ for $\sigma > 0$. Hence $|L(s,\chi)| \le q|s|\int_1^\infty t^{-\sigma-1}dt = q|s|/\sigma$. □

Lemma 2.3. *Let $A_0 = \frac{3}{2}\sum_\ell \ell^{-3/2}$, the sum over primes. For every Dirichlet character χ the series $E(s) = \sum_\ell -\log(1-\chi(\ell)\ell^{-s})$ is holomorphic on $\operatorname{Re} s > 1$, $e^{E(s)} = L(s,\chi)$ there, and $|E(s)| \le A_0$ for $\operatorname{Re} s \ge 3/2$.*

Proof. For $\operatorname{Re} s \ge 1$ we have $|\chi(\ell)\ell^{-s}| \le 1/2$, and $|\log(1-w)| \le \frac{3}{2}|w|$ for $|w| \le 1/2$. The series therefore converges locally uniformly on $\operatorname{Re} s > 1$ and is bounded by A_0 on $\operatorname{Re} s \ge 3/2$, and the Euler product gives $e^E = L$. □

Lemma 2.4. *Let $0 < \delta < 1/1000$, $1/2 \le a \le 1$, and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\delta$. Put $K_1 = (1+2A_0)/(2\delta^2)$. For $a + 8\delta \le \sigma \le 2$ and $t \in \mathbb{R}$,*

$$\Big|\frac{L'}{L}(\sigma + it, \chi)\Big| \le K_1 \log\big(2q(3+|t|)^2\big).$$

Proof. Let $c = 2+it$, $R = 2-a-3\delta$, $r = 2-a-6\delta$ and $\Lambda_0 = \log(2q(3+|t|)^2) \ge \log 18 > 1$. On the disc $|z-c| < 2-a-2\delta$ we have $\operatorname{Re} z > a+2\delta$, so $L(z,\chi)$ has no zero there and has a holomorphic logarithm h. On $|z-c| < 1/2$ both h and E of Lemma 2.3 are logarithms of L, so they differ by a constant in $2\pi i\mathbb{Z}$; we choose $h = E$ there. Then $|h(c)| \le A_0$.

If $|z-c| \le R$, then $\operatorname{Re} z \ge a + 3\delta \ge 1/2$ and $|z| \le |c| + 3/2 \le 4 + |t|$, so Lemma 2.2 gives $\operatorname{Re} h(z) = \log|L(z,\chi)| \le \log(2q(4+|t|)) \le \Lambda_0$. The Borel–Carathéodory inequality [26, §5.5] gives, for $|z-c| \le r$,

$$|h(z) - h(c)| \le \frac{2r}{R-r} \max_{|w-c|=R} \operatorname{Re}\big(h(w) - h(c)\big) \le \frac{\Lambda_0 + A_0}{\delta},$$

as $r \le 3/2$ and $R - r = 3\delta$. Hence $|h(z)| \le A_0 + (\Lambda_0 + A_0)/\delta \le (1+2A_0)\Lambda_0/\delta$ on $|z-c| \le r$. For $w = \sigma + it$ with $a + 8\delta \le \sigma \le 2$ we have $|w - c| = 2 - \sigma \le r - 2\delta$, so the circle $|z - w| = 2\delta$ lies in $|z - c| \le r$, and Cauchy's estimate gives $|L'/L(w,\chi)| = |h'(w)| \le (1+2A_0)\Lambda_0/(2\delta^2)$. □

Lemma 2.5. *For $q \ge 1$, $t \in \mathbb{R}$ and $s = \sigma + it$ with $\sigma \ge 1/2$,*

$$\log\big(2q(3+|t|)^2\big) \le 10\log(q+2)(1+|t|)^{1/2}, \qquad |s(s+1)| \ge \tfrac{1}{8}(1+|t|)^2.$$

Proof. We have $\log(2q) \le 2\log(q+2)$, and $\log y \le 2y^{1/2}$ with $3 + |t| \le 4(1+|t|)$ gives $\log(3+|t|) \le 4(1+|t|)^{1/2}$; the first bound follows from $\log(q+2) \ge 1$. For the second, $|s| \ge \max(\sigma, |t|) \ge (1+|t|)/4$ and $|s+1| \ge \max(\sigma+1, |t|) \ge (1+|t|)/2$. □

Proposition 2.6. *Let $0 < \delta < 1/1000$. There is $K = K(\delta)$ such that the following holds. Let $1/2 \le a \le 1$ and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\delta$. Then for $x \ge 1$,*

$$|S(\chi; x)| \le K\, x^{a+8\delta} \log(q+2).$$

Proof. Put $\sigma_0 = a + 8\delta$ and

$$\Phi(s) = -\frac{L'}{L}(s,\chi)\,\frac{x^s}{s(s+1)}.$$

For $\operatorname{Re} s > 1$, $-L'/L(s,\chi) = \sum_n \chi(n)\Lambda(n)n^{-s}$, so Lemma 2.1 gives $S(\chi;x) = \frac{1}{2\pi}\int \Phi(2+it)\,dt$. The function Φ is holomorphic on $\sigma_0 \le \operatorname{Re} s \le 2$,

and by Lemmas 2.4 and 2.5,

$$|\Phi(\sigma+it)| \le 80K_1 \log(q+2)\, x^{\sigma}(1+|t|)^{-3/2} \qquad (\sigma_0 \le \sigma \le 2).$$

This tends to 0 as $|t| \to \infty$ uniformly in σ, so Cauchy's theorem on the rectangles $[\sigma_0, 2] \times [-T, T]$, $T \to \infty$, moves the integral to $\operatorname{Re} s = \sigma_0$. Since $\int_{\mathbb{R}} (1+|t|)^{-3/2} dt = 4$, we get $|S(\chi; x)| \le \frac{160K_1}{\pi} x^{\sigma_0} \log(q+2)$. □

3. Primitive roots as a combination of characters

Fix a prime p, put $N = p-1$, let g generate $(\mathbb{Z}/p\mathbb{Z})^{\times}$, and let χ_1 be the character modulo p with $\chi_1(g) = e^{2\pi i/N}$. Let χ_0 be the principal character.

Lemma 3.1. *For every integer n,*

$$\mathbf{1}[n \text{ is a primitive root mod } p] = \sum_{d \mid N} \frac{\mu(d)}{d} \sum_{m=0}^{d-1} \chi_1^{mN/d}(n),$$

and $\chi_1^{mN/d} \ne \chi_0$ for $d \mid N$ and $0 < m < d$.

Proof. If $p \mid n$ both sides vanish. Otherwise $n \equiv g^k$, the order of n is $N/(N,k)$, and $\sum_{m<d} e^{2\pi i k m/d} = d\,\mathbf{1}[d \mid k]$; the right side is $\sum_{d \mid (N,k)} \mu(d) = \mathbf{1}[(N,k)=1]$. The character $\chi_1^{mN/d}$ sends g to $e^{2\pi i m/d} \ne 1$. □

Lemma 3.2. *For $N \ge 1$, $\sum_{d \mid N} \mu(d)/d = \varphi(N)/N \ge 1/\tau(N)$.*

Proof. $\varphi(N)/N = \prod_{r \mid N}(1-1/r) \ge 2^{-\omega(N)} \ge 1/\tau(N)$, the product over the prime divisors r of N. □

Lemma 3.3. *For $x \ge 10^4$, $\operatorname{Re} S(\chi_0; x) \ge \frac{1}{2}\psi(x/2) - \log x \ge x/10$.*

Proof. The terms of $\sum_{n \le x} \Lambda(n)(1-n/x)$ are nonnegative and at least $\Lambda(n)/2$ for $n \le x/2$, so the sum is at least $\psi(x/2)/2$. The terms removed by χ_0 are those with $p \mid n$, that is $n = p^k \le x$, and they contribute at most $\sum_{p^k \le x} \log p \le \log x$. For the second inequality, $\psi(n) \ge n \log 2$ for integers $n \ge 7$ [18], so $\psi(x/2) \ge (x/2-1)\log 2$, and $\frac{x}{4}\log 2 - \frac{1}{2}\log 2 - \log x \ge \frac{x}{10}$ for $x \ge 10^4$. □

Lemma 3.4. *If ℓ^k with $k \ge 1$ is a primitive root modulo p, so is ℓ.*

Proof. ℓ is a unit modulo p, and $N = \operatorname{ord}(\ell^k)$ divides $\operatorname{ord}(\ell)$, which divides N. □

4. Proof of Theorem 1.1 and Corollaries 1.2 and 1.3

Proof of Theorem 1.1. Let $\delta = \eta/(2000(8+\eta))$, so $0 < \delta < 1/1000$, and let K be given by Proposition 2.6; put $K' = \max(K, 1)$. Let $\sigma_0 = 7/8 + 8\delta$ and $\beta = 1 - \sigma_0 = 1/8 - 8\delta > 0$. Then

$$(8+\eta)\beta = 1 + \frac{\eta}{8} - \frac{\eta}{250} \ge 1.$$

Put $C_\eta = \max\bigl(10^4, (20K')^{8+\eta}\bigr)$. Given a prime p, let $\tau = \tau(p-1)$, $Y = \tau^2\log(p+2) \ge 1$ and $x = C_\eta Y^{8+\eta} \ge 10^4$.

By Theorem 1.8, Proposition 2.6 applies with $a = 7/8$ to every nonprincipal χ modulo p, so $|S(\chi;x)| \le B := K'x^{\sigma_0}\log(p+2)$. Let G be the sum $S(f;x)$ with f the indicator of the primitive roots modulo p. By Lemma 3.1,

$$G = c_0 S(\chi_0;x) + \mathcal{E}, \qquad c_0 = \sum_{d\mid N}\frac{\mu(d)}{d}, \qquad \mathcal{E} = \sum_{d\mid N}\frac{\mu(d)}{d}\sum_{0<m<d} S(\chi_1^{mN/d};x),$$

and $|\mathcal{E}| \le \sum_{d\mid N}\frac{d-1}{d}B \le \tau B$. Lemmas 3.2 and 3.3 give

$$|G| \ge c_0 \operatorname{Re} S(\chi_0;x) - |\mathcal{E}| \ge \frac{x}{10\tau} - \tau B.$$

Now $x^\beta = C_\eta^\beta Y^{(8+\eta)\beta} \ge (20K')^{(8+\eta)\beta}Y \ge 20K'Y$, so $\tau^2 B = K'Yx^{\sigma_0} \le x^{\beta+\sigma_0}/20 = x/20$. Hence $|G| \ge x/(20\tau) > 0$, and some $n \le x$ with $\Lambda(n) \ne 0$ is a primitive root modulo p. Writing $n = \ell^k$, the prime $\ell \le x$ is a primitive root by Lemma 3.4. □

Proof of Corollary 1.2. Take $\eta = 1$ and $\nu = \varepsilon/27$. By [13, Thm. 315] there is D_ν with $\tau(n) \le D_\nu n^\nu$, and $\log(p+2) \le (p+2)^\nu/\nu \le 3^\nu p^\nu/\nu$. Theorem 1.1 gives $\ell \le C_1(D_\nu^2 3^\nu \nu^{-1})^9 p^{27\nu}$. □

Proof of Corollary 1.3. Corollary 1.2 with $\varepsilon = 1/2$ gives $\ell \le C\sqrt{p} < p$ for $p > C^2$. □

5. The Lindelöf bound and character sums

In this section A_0 is the constant of Lemma 2.3.

Lemma 5.1. *Let $0 < \kappa < 1/1000$ and $\varepsilon > 0$. There is $D = D(\kappa,\varepsilon)$ such that the following holds. Let $1/2 \le a \le 1$ and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. For $a + 8\kappa \le \sigma \le 2$ and $t \in \mathbb{R}$,*

$$|L(\sigma+it,\chi)| + |L(\sigma+it,\chi)|^{-1} \le D\bigl(2q(3+|t|)^2\bigr)^{\varepsilon}.$$

Proof. Let c, r, Λ_0 and the logarithm h of $L(z,\chi)$ on $|z-c| < 2-a-2\kappa$ be as in the proof of Lemma 2.4, with δ replaced by κ. Both h and E are logarithms of L on the disc $|z-c| < 3/4$, where $\operatorname{Re} z > 1$, so $h = E$ there, and $|h(z)| \le A_0$ for $|z-c| \le 1/2$ by Lemma 2.3. The proof of Lemma 2.4 gives $|h(z)| \le M := (1+2A_0)\Lambda_0/\kappa$ for $|z-c| \le r$.

Let $w = \sigma+it$, $\rho = |w-c| = 2-\sigma$, $A_1 = \max(A_0,1)$ and $\lambda = 4\kappa/(3\log 3)$; note $M \ge 1$. If $\rho \le 1/2$ then $|h(w)| \le A_0 \le A_1 M^{1-\lambda}$. Otherwise $1/2 < \rho \le r - 2\kappa$, and Hadamard's three circles theorem [26, §5.3] gives $|h(w)| \le A_0^{1-\alpha}M^\alpha$ with $\alpha = \log(2\rho)/\log(2r)$. Since $1 - 6\kappa \le r \le 3/2$,

$$\alpha \le \frac{\log(2r-4\kappa)}{\log(2r)} = 1 + \frac{\log(1-2\kappa/r)}{\log(2r)} \le 1 - \frac{2\kappa/r}{\log 3} \le 1 - \lambda,$$

so again $|h(w)| \le A_1 M^{1-\lambda} = B\Lambda_0^{1-\lambda}$ with $B = A_1((1+2A_0)/\kappa)^{1-\lambda}$. If $\Lambda_0^\lambda \ge B/\varepsilon$ then $B\Lambda_0^{1-\lambda} \le \varepsilon\Lambda_0$, and otherwise $B\Lambda_0^{1-\lambda} \le D_1 := B(B/\varepsilon)^{(1-\lambda)/\lambda}$. Since $\log|L(w,\chi)| = \operatorname{Re} h(w)$,

$$|L(w,\chi)|^{\pm 1} \le e^{|h(w)|} \le e^{D_1} e^{\varepsilon\Lambda_0} = e^{D_1}\big(2q(3+|t|)^2\big)^{\varepsilon},$$

and $D = 2e^{D_1}$ works. □

Lemma 5.2. *Let $|b(n)| \le 1$ for $n \ge 1$ and put $T(y) = \sum_{n\le y} b(n)(y-n)$ for $y > 0$. For $x, h > 0$,*

$$h\Big|\sum_{n\le x} b(n)\Big| \le |T(x+h) - T(x)| + h(h+1).$$

Proof. $T(x+h) - T(x) = h\sum_{n\le x} b(n) + \sum_{x<n\le x+h} b(n)(x+h-n)$, and the last sum has at most $h+1$ terms, each of modulus at most h. □

Proposition 5.3. *Let $0 < \kappa < 1/1000$ and $0 < \varepsilon \le 1/8$. There is $K = K(\kappa,\varepsilon)$ such that the following holds. Let $1/2 \le a$ with $a + 8\kappa \le 1$, and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. Then for $x \ge 1$,*

$$\Big|\sum_{n\le x} \chi(n)\Big| \le K\, q^{\varepsilon} x^{a+8\kappa+4\varepsilon}.$$

Proof. Put $\sigma_0 = a + 8\kappa \in [1/2, 1]$ and let D be given by Lemma 5.1. Since $2q(3+|t|)^2 \le 18q(1+|t|)^2$, Lemmas 5.1 and 2.5 give, for $s = \sigma + it$ with $\sigma_0 \le \sigma \le 2$,

$$\Big|\frac{L(s,\chi)}{s(s+1)}\Big| \le 8D(18q)^{\varepsilon}(1+|t|)^{-2+2\varepsilon}. \tag{1}$$

Let $T(y) = \sum_{n\le y}\chi(n)(y-n)$. Lemma 2.1 with $a(n) = \chi(n)$ gives $T(y) = \frac{1}{2\pi i}\int_{(2)} L(s,\chi)\, y^{s+1}\frac{ds}{s(s+1)}$. The integrand is entire, and by (1) it tends to 0 as $|t| \to \infty$ uniformly in $\sigma_0 \le \sigma \le 2$, since $2\varepsilon < 1$; Cauchy's theorem moves the line to $\operatorname{Re} s = \sigma_0$. For $h > 0$,

$$T(x+h) - T(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{L(s,\chi)}{s(s+1)}\big((x+h)^{s+1} - x^{s+1}\big)\, dt, \qquad s = \sigma_0 + it.$$

Since $(x+h)^{s+1} - x^{s+1} = (s+1)\int_x^{x+h} u^s\, du$ and $|s+1| \le 3(1+|t|)$, its modulus is at most $3(1+|t|)h(x+h)^{\sigma_0}$, and also at most $2(x+h)^{\sigma_0+1}$. The minimum of two numbers is at most the product of their powers $1-4\varepsilon$ and 4ε, so

$$\big|(x+h)^{s+1} - x^{s+1}\big| \le (1+|t|)^{1-4\varepsilon}(x+h)^{\sigma_0}(3h)^{1-4\varepsilon}\big(2(x+h)\big)^{4\varepsilon}.$$

With (1) and $\int_{\mathbb{R}}(1+|t|)^{-1-2\varepsilon}dt = 1/\varepsilon$,

$$|T(x+h) - T(x)| \le \frac{4D(18q)^{\varepsilon}}{\pi\varepsilon}(x+h)^{\sigma_0}(3h)^{1-4\varepsilon}\big(2(x+h)\big)^{4\varepsilon}.$$

Take $h = x^{\sigma_0}$, so $1 \le h \le x$. Then $(x+h)^{\sigma_0} \le 2h$, $(3h)^{1-4\varepsilon} \le 3h$ and $(2(x+h))^{4\varepsilon} \le 4x^{4\varepsilon}$, so the right side is at most $\frac{1728D}{\pi\varepsilon} q^{\varepsilon} h^2 x^{4\varepsilon}$. By Lemma 5.2 with $b = \chi$,

$$\Big|\sum_{n\le x} \chi(n)\Big| \le \frac{1728D}{\pi\varepsilon} q^{\varepsilon} h x^{4\varepsilon} + h + 1 \le \Big(\frac{1728D}{\pi\varepsilon} + 2\Big) q^{\varepsilon} x^{\sigma_0+4\varepsilon}. \qquad \square$$

Proof of Theorem 1.4. Let $\kappa = \min(\delta/8, 1/2000)$ and $a = 7/8 - 2\kappa$, so $1/2 \le a \le 1$ and, by Theorem 1.8, $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. Let D be given by Lemma 5.1 with $\varepsilon = \delta/2$. If $7/8+\delta \le \sigma \le 2$, then $\sigma \ge a + 8\kappa$, and $2q(3+|t|)^2 \le (2q(3+|t|))^2$ gives $|L| + |L|^{-1} \le D 2^{\delta} (q(3+|t|))^{\delta}$. If $\sigma > 2$, Lemma 2.3 gives $|L|^{\pm 1} \le e^{A_0}$. So $C_\delta = D2^\delta + 2e^{A_0}$ works. $\square$

Proof of Theorem 1.5. Let $\delta' = \min(\delta, 1)$, $\kappa = \min(\delta'/12, 1/2000)$, $a = 7/8 - 2\kappa$ and $\varepsilon = \delta'/8$. Then $a + 8\kappa = 7/8 + 6\kappa \le 1$, and Proposition 5.3, with Theorem 1.8, gives the bound $K q^{\delta'/8} x^{7/8+6\kappa+\delta'/2} \le K q^{\delta} x^{7/8+\delta}$. $\square$

Proof of Corollary 1.6. Let $\delta = (\theta - 7/8)\eta/(1+\eta)$, so that $\delta/\eta + 7/8 + \delta = \theta$. For $x \ge q^{\eta}$ we have $q^{\delta} \le x^{\delta/\eta}$, and Theorem 1.5 gives $|\sum_{n\le x} \chi(n)| \le C_\delta x^{\theta}$. With $\theta = 15/16$ this is $o(x)$. $\square$

6. Exponential sums with multiplicative coefficients

Write $e(x) = e^{2\pi i x}$ and let $\|x\|$ be the distance from x to the nearest integer. For $M > 0$ put $w_M(\theta) = M$ if $\|\theta\| < 1/M$ and $w_M(\theta) = 1/\|\theta\|$ otherwise; then $0 \le w_M(\theta) \le M$ and $w_M(-\theta) = w_M(\theta)$.

Theorem 6.1. *There is an absolute constant C_1 with the following property. Let f be completely multiplicative with $|f(n)| \le 1$, let $\alpha \in \mathbb{R}$, and let a, r be coprime integers with $r \ge 1$, $|\alpha - a/r| \le r^{-2}$ and $2 \le R \le r \le N/R$. Then*

$$\Big|\sum_{n\le N} f(n) e(n\alpha)\Big| \le C_1 \Big(\frac{N}{\log N} + \frac{N(\log N)^2}{R^{1/4}}\Big).$$

Montgomery and Vaughan [17, Corollary 1] proved this for multiplicative f with $N(\log R)^{3/2} R^{-1/2}$ in place of $N(\log N)^2 R^{-1/4}$. The proof below follows their approach (a bilinear decomposition, the spacing of the multiples of α and an upper bound for prime pairs) with cruder estimates, which suffice for Theorem 1.7.

Lemma 6.2. *Let a, r be coprime with $r \ge 2$ and $|\alpha - a/r| \le r^{-2}$, and let $H \ge 0$ be an integer.*

(i) $\|k\alpha\| \ge 1/(2r)$ *for every integer k with $0 < |k| \le r/2$.*
(ii) $\#\{1 \le h \le H : \|h\alpha\| < t\} \le (4H/r + 1)(4rt + 1)$ *for every $t \ge 0$.*

(iii) *For $M > 0$ and every integer $K \geq 0$ with $M \leq 2^K$,*

$$\sum_{h\leq H} w_M(h\alpha) \leq \Big(\frac{4H}{r}+1\Big)(3M+4r+8rK),$$

$$\sum_{h\leq H} w_M(h\alpha)^2 \leq \Big(\frac{4H}{r}+1\Big)(20rM+3M^2).$$

Proof. (i) Let z be the integer nearest to $k\alpha$. Since $0 < |k| < r$ and $(a,r)=1$, $r \nmid ka$, so $|ka/r - z| \geq 1/r$ and $\|k\alpha\| \geq 1/r - |k|\,|\alpha - a/r| \geq 1/(2r)$.

(ii) Let $L = \lfloor r/2 \rfloor$, so $L \geq r/4$. For k in a block of L consecutive integers let $y_k = k\alpha - z_k$ with z_k the integer nearest to $k\alpha$. For $k \neq k'$ in the block, $0 < |k-k'| < L \leq r/2$ and $y_k - y_{k'} \equiv (k-k')\alpha \pmod 1$, so $|y_k - y_{k'}| \geq 1/(2r)$ by (i). Points of $(-t,t)$ at mutual distances at least $1/(2r)$ number at most $4rt+1$, and $[1,H]$ is covered by at most $H/L + 1 \leq 4H/r + 1$ blocks.

(iii) If $\|\theta\| \geq 1/M$, let $k \geq 1$ be least with $\|\theta\| < 2^k/M$; then $k \leq K$ because $\|\theta\| \leq 1/2 < 2^K/M$, and $\|\theta\| \geq 2^{k-1}/M$, so $w_M(\theta) \leq 2^{1-k}M$. Hence

$$w_M(\theta)^j \leq M^j\,[\|\theta\| < 1/M] + \sum_{k=1}^{K} (2^{1-k}M)^j\,[\|\theta\| < 2^k/M] \qquad (j = 1, 2).$$

Summing over $h \leq H$ with (ii), the first sum is at most $(4H/r+1)\big(4r + M + \sum_{k\leq K}(8r + 2^{1-k}M)\big)$ and the second at most $(4H/r+1)\big(4rM + M^2 + \sum_{k\geq 1}(16rM2^{-k} + 4M^2 4^{-k})\big)$. □

Lemma 6.3. *Let $\theta, \alpha \in \mathbb{R}$, $U > 0$, and let A, B, N be nonnegative integers with $B \leq A + U$.*

(i) $\big|\sum_{A<x\leq B} e(x\theta)\big| \leq w_U(\theta)$.

(ii) *For a finite set V of positive integers and $b : V \to \mathbb{C}$,*

$$\sum_{A<u\leq B} \Big|\sum_{\substack{v\in V\\ uv\leq N}} b(v)e(uv\alpha)\Big|^2 \leq \sum_{v,v'\in V} |b(v)|\,|b(v')|\,w_U\big((v-v')\alpha\big).$$

(iii) *For $M > 0$ and an integer $H \geq 0$, $\sum_{1\leq m,m'\leq H} w_M((m-m')\alpha) \leq H\big(M + 2\sum_{h\leq H} w_M(h\alpha)\big)$.*

Proof. (i) If $\|\theta\| < 1/U$, the sum has at most U terms of modulus 1. Otherwise $\theta \notin \mathbb{Z}$, the sum has modulus at most $|e((B-A)\theta) - 1|/|e(\theta) - 1|$, and $|e(\theta) - 1| = 2|\sin \pi\theta| \geq 4\|\theta\|$, so the modulus is at most $1/(2\|\theta\|)$. (ii) Expanding the square gives $\sum_{v,v'} b(v)\overline{b(v')} \sum_u e(u(v-v')\alpha)$, where u runs over the integers of $(A, \min(B, N/v, N/v')]$, an interval of length at most U; apply (i). (iii) For fixed m, the term $m' = m$ is $w_M(0) = M$, and each $h = |m - m'| \in [1, H]$ arises from at most two m'. □

Lemma 6.4. *Let $C_5 = 1 + 4\zeta(3/2)$. For every completely multiplicative f with $|f(n)| \le 1$, every $\alpha \in \mathbb{R}$ and every integer $N \ge 1$, the sum $S = \sum_{n\le N} f(n)e(n\alpha)$ satisfies*

$$\Big|S\log N - \sum_{p\le N} \log p\, f(p) \sum_{m\le N/p} f(m)e(pm\alpha)\Big| \le C_5 N.$$

Proof. Write $S\log N = S_1 + S_2$ with $S_1 = \sum_{n\le N} f(n)e(n\alpha)\log n$ and $S_2 = \sum_{n\le N} f(n)e(n\alpha)\log(N/n)$. Then S_2 has modulus at most $N\log N - \log N! \le N$, since $N^N/N! \le e^N$. In S_1 write $\log n = \sum_{d|n}\Lambda(d)$ and $n = dm$; complete multiplicativity gives $\sum_{d\le N}\Lambda(d)f(d)\sum_{m\le N/d} f(m)e(dm\alpha)$. The terms with d prime form the sum in the statement, and the others have modulus at most $N\sum_p\sum_{k\ge 2}\log p/p^k \le 2N\sum_p \log p/p^2 \le 4\zeta(3/2)N$, since $\log p \le 2\sqrt{p}$. □

Let $G(h) = \prod_{p|h} p/(p-1)$.

Lemma 6.5. (i) *For every integer $X \ge 2$ and every even $h \ne 0$,*

$$\#\{0 \le n < X : n \text{ and } n+h \text{ prime}\} \le \frac{400\,G(h)^2 X}{(\log X)^2} + 3X^{3/5}.$$

(ii) $\sum_{h\le X} G(h)^4 \le e^{30}X$ *for every $X \ge 1$.*

Proof. (i) We use Selberg's upper bound sieve [12, Ch. 3] in the following form. Let $a_0, \dots, a_{X-1} \ge 0$ be integers, P a squarefree integer, ν a multiplicative function with $0 < \nu(p) < 1$ for $p \mid P$, $y \ge 1$, and $R_d = \#\{n : d \mid a_n\} - \nu(d)X$. Then

$$\#\{n : (a_n, P) = 1\} \le \frac{X}{S} + \sum_{\substack{d\mid P\\ d\le y}} 3^{\omega(d)}|R_d|, \qquad S = \sum_{\substack{l\mid P\\ l^2\le y}} \prod_{p\mid l} \frac{\nu(p)}{1-\nu(p)}.$$

Take $a_n = n(n+h)$, $y = X^{1/5}$, $z = \lfloor y\rfloor$, P the product of the primes $p \le z$ with $p \nmid h$, and $\nu(d) = 2^{\Omega(d)}/d$. Every $p \mid P$ is odd, so $\nu(p) = 2/p < 1$. For $d \mid P$ the residues n with $d \mid n(n+h)$ form $2^{\omega(d)} = \tau(d)$ classes modulo d (namely $n \equiv 0$ or $-h \pmod p$ for each $p \mid d$, distinct as $p \nmid h$), and each class contains between $X/d - 1$ and $X/d + 1$ of the integers $0 \le n < X$; since $\nu(d)X = \tau(d)X/d$, $|R_d| \le \tau(d)$. Since the prime factors of d are at least 3, $3^{\omega(d)}|R_d| \le d^2 \le y^2$, and the error term is at most $y^3 = X^{3/5}$.

Since ν is completely multiplicative, $\prod_{p|l}\nu(p)/(1-\nu(p))$ is the sum of $\nu(m)$ over the m whose set of prime factors is that of l; so $S \ge \sum \nu(m)$ over the $m \le \sqrt{y}$ whose prime factors divide P. Let $T = \lfloor\lfloor\sqrt{y}\rfloor^{1/2}\rfloor$ and let B be the set of $a \le T$ with no prime factor dividing h. For $a, b \in B$ the product $m = ab \le \sqrt{y}$ has its prime factors in P, and $\nu(m) \ge \tau(m)/m$, so $S \ge \big(\sum_{a\in B} 1/a\big)^2$. Removing the multiples of one prime p from a sum

$\sum 1/a$ over $a \le T$ divides it by at most $\sum_k p^{-k} = p/(p-1)$, since each a is $p^k a'$ with $p \nmid a'$; hence $\sum_{a\in B} 1/a \ge G(h)^{-1} \sum_{a\le T} 1/a \ge G(h)^{-1} \log(T+1)$. As $(T+1)^2 > \sqrt{y}$, $\log(T+1) \ge \log X/20$, and $X/S \le 400 G(h)^2 X/(\log X)^2$. Finally, if $n < X$ and $n, n+h$ are prime, then $n \le z$ or $(n(n+h), P) = 1$, so the count is at most $z + 1 + X/S + y^3 \le 400 G(h)^2 X/(\log X)^2 + 3X^{3/5}$.

(ii) For $t = p-1 \ge 1$, $(t+1)^5 \le (t+31)t^4$, that is, $(p/(p-1))^4 \le 1 + 30/p$. Hence $G(h)^4 \le \sum_d 30^{\omega(d)}/d$ over the squarefree $d \mid h$, and

$$\sum_{h\le X} G(h)^4 \le \sum_{d\le X} \mu(d)^2 \frac{30^{\omega(d)}}{d} \cdot \frac{X}{d} \le X \prod_p \Big(1 + \frac{30}{p^2}\Big)$$
$$\le X \exp\Big(30 \sum_{n\ge 2} \frac{1}{n^2}\Big) \le e^{30} X. \qquad \square$$

Proof of Theorem 6.1. Let $S = \sum_{n\le N} f(n) e(n\alpha)$, $L = \log N$ and $\ell = \lfloor \log_2 N \rfloor$. Since $N \ge rR \ge 4$, $L \ge 1$. If $\ell < 64$, then $L < 64 \log 2 < 45$ and $|S| \le N \le 45N/L$. Let $\ell \ge 64$, $I = \lfloor \log_2 \ell \rfloor + 1$ and $J = \ell + 1 - 4I$. Then $16^I > \ell^4$, $1 \le J \le \ell \le 2L$, $I \le \ell \le 2L$, $2^I \le 2\ell$, $N < 2^{\ell+1} = 2^J 16^I$ and $J^2 2^J \le (\ell+1)^2 2^{\ell+1} \ell^{-4} \le 2N$. By Lemma 6.4, $|S|L \le C_5 N + |T_1| + |T_2|$, where T_1 and T_2 are the parts of $\sum_p \log p\, f(p) \sum_{m\le N/p} f(m) e(pm\alpha)$ with $p \le 2^J$ and with $p > 2^J$.

The small primes. Let $0 \le j < J$, $U = 2^j$, $H = \lfloor N/U \rfloor$, and for $U < x \le 2U$ let $B(x) = \sum_{m\le H,\, xm\le N} f(m) e(xm\alpha)$; the inner sum for a prime $p \in (U, 2U]$ is $B(p)$. By Cauchy–Schwarz, Lemma 6.3(ii), (iii) and Lemma 6.2(iii) with $M = U$, $K = j$,

$$\sum_{U<p\le 2U} \log p\, |B(p)| \le \big(U \log^2(2U)\big)^{1/2} Q_j^{1/2},$$

where $Q_j = H\big(U + 2(4H/r + 1)(3U + 4r + 8rj)\big)$. Since $HU \le N$, $UQ_j \le 7NU + (8 + 16j)Nr + 24N^2/r + (32 + 64j)N^2/U$, and $\log(2U) \le j + 1$. Summing over $j < J$ with $j+1 \le J$, $\sum_{j<J} 2^{j/2} \le 3 \cdot 2^{J/2}$, $\sqrt{(32 + 64j)2^{-j}} \le 8(j+1)(3/4)^j$ and $\sum_{j\ge 0} (j+1)^2 (3/4)^j = 112$,

$$|T_1| \le 3\sqrt{7} J \sqrt{N 2^J} + J^2 N \sqrt{24/r} + 896N + J^2 \sqrt{24 JNr}.$$

Now $J^2 2^J \le 2N$ gives $3\sqrt{7} J \sqrt{N 2^J} \le 12N$; $r \ge R \ge \sqrt{R}$ gives $\sqrt{24/r} \le 5R^{-1/4}$; and $r\sqrt{R} \le rR \le N$ gives $\sqrt{24JNr} \le 5JNR^{-1/4}$. With $J \le 2L$ and $L \ge 1$, $|T_1| \le 908N + 60L^3 NR^{-1/4}$.

The large primes. For $p > 2^J$ we have $N/p < 16^I$, so the m in T_2 lie in $(0, 16^I - 1]$, the union of the blocks $(16^i - 1, 16^{i+1} - 1]$ with $0 \le i < I$. Fix i, let $A = 16^i - 1$, $U = 15 \cdot 16^i$ and $X = \lfloor N/16^i \rfloor$, and assume $X \ge 1$, the block being empty otherwise. Only the primes $p \in (2^J, X]$ contribute, since $p \ge X + 1$ and $m \ge 16^i$ give $pm > N$. Let V be the set of these primes, which are odd, and $B(m) = \sum_{p\in V,\, pm\le N} \log p\, f(p) e(pm\alpha)$. The

block is $\sum_{A<m\le A+U} f(m)B(m)$, and by Cauchy–Schwarz, Lemma 6.3(ii) and $\log p \le \lambda = \log(X+1)$, its square is at most $U\lambda^2 W$, where

$$W = \sum_{p,p'\in V} w_U((p-p')\alpha) = |V|U + 2\sum_{h=1}^{X} w_U(h\alpha)\,\#\{p \in V : p+h \in V\}.$$

The count vanishes for odd h, and for even h Lemma 6.5(i), applied to $X+1$, bounds it by $400G(h)^2(X+1)\lambda^{-2}+3(X+1)^{3/5}$. By Cauchy–Schwarz, Lemma 6.5(ii) and Lemma 6.2(iii), $\sum_{h\le X} G(h)^2 w_U(h\alpha) \le e^{15}\sqrt{X}Q^{1/2}$ with $Q = (4X/r+1)(20rU+3U^2)$, and $\sum_{h\le X} w_U(h\alpha) \le XU$. Hence $U\lambda^2 W \le A_1 + 800e^{15}Z + A_3$, where

$$A_1 = U^2\lambda^2 X, \qquad Z = U(X+1)\sqrt{X}Q^{1/2}, \qquad A_3 = 6U^2\lambda^2(X+1)^{3/5}X.$$

Let $\kappa = \log(N+1) \ge \lambda$ and $T_i = 4^i\kappa\,(N(2N)^{3/5})^{1/2}$. Since $U^2X = 225\cdot 16^i(16^iX) \le 225\cdot 16^i N$ and $X+1 \le 2N$, we get $A_1 \le 225T_i^2$ and $A_3 \le 1350T_i^2 \le 37^2T_i^2$. With $w = UX \le 15N$, $X \le N/16^i$ and $R \le r \le N/R$,

$$\begin{aligned} Z^2 \le 4U^2X^3Q &= 320w^3X + \frac{48w^4}{r} + 80rw^3 + 12Uw^3 \\ &\le \frac{320\cdot 15^3N^4}{16^i} + \frac{48\cdot 15^4N^4}{R} + \frac{80\cdot 15^3N^4}{R} \\ &\quad + 180\cdot 15^3\cdot 16^iN^3, \end{aligned}$$

and every coefficient is at most 40^4, so $Z^{1/2} \le 40(N2^{-i}+2NR^{-1/4}+2^iN^{3/4})$. The block is therefore at most

$$\begin{aligned} &15T_i + 40\sqrt{800e^{15}}\big(N2^{-i} + 2NR^{-1/4} + 2^iN^{3/4}\big) + 37T_i \\ &\quad \le K_b\big(N2^{-i} + NR^{-1/4} + 2^iN^{3/4} + T_i\big), \end{aligned}$$

with $K_b = 80\sqrt{800e^{15}} + 52$. Summing over $i < I$,

$$|T_2| \le K_b\big(2N + INR^{-1/4} + 2^IN^{3/4} + 4^I\kappa(N(2N)^{3/5})^{1/2}\big).$$

Here $I \le 2L^3$; $2^I \le 4L \le 16N^{1/4}$; $4^I \le 16L^2$ and $\kappa \le 2L$; and $\log x \le 15x^{1/15}$ gives $L^3(N(2N)^{3/5})^{1/2} \le 2\cdot 15^3N$. Hence $|T_2| \le K_b(216018N + 2L^3NR^{-1/4})$.

Altogether $|S|L \le (C_5 + 908 + 216018K_b)N + (60+2K_b)L^3NR^{-1/4}$, and dividing by L gives the theorem with $C_1 = 45 + C_5 + 908 + 216018K_b + 60 + 2K_b$. □

7. The Pólya–Vinogradov inequality

Let $P(n)$ be the largest prime factor of $n \ge 2$, with $P(1) = 1$. In this section χ is a primitive character modulo $q \ge 3$.

Lemma 7.1. *Let $B \geq 0$ be such that $\left|\sum_{m\leq H} \bar{\chi}(m)e(m\beta)/m\right| \leq B$ for every integer $H \leq q/2$ and every $\beta \in \mathbb{R}$. Then $\left|\sum_{n\leq T} \chi(n)\right| \leq \sqrt{q}\,(B+6)$ for every $T \geq 0$.*

Proof. Let $\tau = \sum_{a \bmod q} \bar{\chi}(a)e(a/q)$. Since $\bar{\chi}$ is primitive, $\sum_a \bar{\chi}(a)e(an/q) = \chi(n)\tau$ for every n, and summing the squares of these identities over n mod q gives $q\varphi(q) = |\tau|^2\varphi(q)$, so $|\tau| = \sqrt{q}$. Summing over $n \leq T$,

$$\tau \sum_{n\leq T} \chi(n) = \sum_{0<|a|\leq q/2} \bar{\chi}(a)\,\frac{e(aT/q)-1}{1-e(-a/q)},$$

where a runs over the integers with $-q/2 < a \leq q/2$, $a \neq 0$. For $0 < |\theta| \leq 1/2$ put $r(\theta) = (1-e(-\theta))^{-1} - (2\pi i\theta)^{-1}$; then $|1-e(-\theta)| = 2|\sin\pi\theta| \geq 4|\theta|$. If $2\pi|\theta| \leq 1$, then $|e(-\theta) - 1 + 2\pi i\theta| \leq (2\pi\theta)^2$ and $|r(\theta)| \leq (2\pi\theta)^2/(4|\theta| \cdot 2\pi|\theta|) < 3$; otherwise $|r(\theta)| \leq 1/(4|\theta|) + 1/(2\pi|\theta|) < 3$. The terms with $r(a/q)$ contribute at most $6q$. Writing $a = \pm m$ with $1 \leq m \leq q/2$, the remaining terms are $q/(2\pi i)$ times a combination, with coefficients of modulus at most 1, of four sums $\sum_{m\leq H} \bar{\chi}(m)e(m\beta)/m$ with $H \leq q/2$, so they contribute at most $2qB/\pi$. Dividing by $|\tau|$ gives the claim. □

Lemma 7.2. *For $N \geq 1$, $\sum_{p\leq N} 1/p \leq \log\log N + 10$. For $Y \geq 2$ and $H \geq 1$,*

$$\sum_{\substack{n\leq H\\ P(n)\leq Y}} \frac{1}{n} \leq e^{11}\log Y.$$

Proof. Legendre's formula gives $\log N! \geq \sum_{p\leq N} \lfloor N/p\rfloor \log p$, and $\log N! \leq N\log N$, $\sum_{p\leq N} \log p \leq N\log 4$; hence $\sum_{p\leq N} \log p/p \leq \log N + \log 4$, and partial summation gives the first bound. The sum in the second is at most $\prod_{p\leq Y}(1-1/p)^{-1} \leq \exp\bigl(\sum_{p\leq Y} \frac{1}{p-1}\bigr)$, and $\sum_p \bigl(\frac{1}{p-1} - \frac{1}{p}\bigr) \leq 1$. □

Lemma 7.3. *There is an absolute constant C_2 such that for $0 < r < q$, $b \in \mathbb{Z}$, $\beta \in \mathbb{R}$, integers $Y \geq \max(2, r)$ and $t \geq 1$,*

$$\Bigl|\sum_{\substack{n\leq t\\ P(n)>Y}} \chi(n)e\Bigl(n\Bigl(\frac{b}{r}+\beta\Bigr)\Bigr)\Bigr| \leq C_2\,\frac{r\log(qr+2)\,(2+2\pi|\beta|t)\,t}{Y^{1/20}\log Y}.$$

Proof. Let ξ be a nonprincipal character modulo k. Proposition 2.6 with $\delta = 1/2000$ and $a = 7/8 - 2\delta$, which applies by Theorem 1.8, gives $|S(\xi;y)| \leq Ky^{7/8+6\delta}\log(k+2)$ for $y \geq 1$. Since $(x+h)S(\xi;x+h) - xS(\xi;x) = h\sum_{n\leq x} \xi(n)\Lambda(n) + \sum_{x<n\leq x+h} \xi(n)\Lambda(n)(x+h-n)$, the choice $h = x^{47/50}$, $\log x \leq 100x^{1/100}$ and the removal of the prime powers give $\bigl|\sum_{p\leq x} \xi(p)\log p\bigr| \leq C_3x^{19/20}\log(k+2)$, and partial summation gives, for

integers $2 \le A \le B$ and $\gamma \in \mathbb{R}$,

$$\Big| \sum_{A<p\le B} \xi(p) e(p\gamma) \Big| \le 2C_3 \, \frac{B^{19/20} \log(k+2)(2+2\pi|\gamma|B)}{\log A}. \tag{2}$$

Every $n \le t$ with $P(n) > Y$ is uniquely $n = pm$ with $p = P(n)$ prime, $Y < p \le t/m$ and $P(m) \le p$, and then $m \le t/Y$. For such p we have $p > r$, so $p \nmid r$, and on the units modulo r the function $u \mapsto e(umb/r)$ equals $\sum_\psi c_\psi \psi(u)$, the sum over the characters ψ modulo r, with $|c_\psi| \le 1$. The character $\chi\psi$ modulo qr is nonprincipal: otherwise $\chi(n) = \bar\psi(n)$ for every n prime to qr, so χ would be induced by a character modulo $\gcd(q,r) < q$. Applying (2) to $\xi = \chi\psi$ with $\gamma = m\beta$, $A = \max(Y, P(m)-1)$ and $B = \lfloor t/m \rfloor$ gives

$$\Big| \sum_{\substack{n\le t \\ P(n)>Y}} \chi(n) e\Big(n\Big(\frac{b}{r}+\beta\Big)\Big) \Big| \le 2C_3 r \, \frac{\log(qr+2)(2+2\pi|\beta|t)}{\log Y} \sum_{m\le t/Y} \Big(\frac{t}{m}\Big)^{19/20},$$

and $\sum_{m\le M} m^{-19/20} \le 20 M^{1/20}$. □

Proposition 7.4. *There is an absolute constant C_4 such that if $\log q \ge 200$, then $\big|\sum_{m\le H} \chi(m) e(m\beta)/m\big| \le C_4 \log\log q$ for every integer $H \le q$ and every $\beta \in \mathbb{R}$.*

Proof. Let $L = \log q$, $M = \lceil L \rceil$, $R = M^{16}$ and $Y = M^{700}$. Since $e^{L/16} \ge (L/16)^4/24 > L+1 \ge M$, we have $2 \le R < q$ and $R \le Y$. Split the sum according to $P(m) \le Y$ or $P(m) > Y$. By Lemma 7.2 the first part is at most $e^{11} \log Y$. For the second put $A(t) = \sum_{n\le t,\, P(n)>Y} \chi(n) e(n\beta)$ and $w(t) = 1/t - 1/(t+1)$. Partial summation and $|A(H)| \le H$ give

$$\Big| \sum_{\substack{m\le H \\ P(m)>Y}} \frac{\chi(m) e(m\beta)}{m} \Big| \le 1 + \sum_{1\le t<H} |A(t)| \, w(t).$$

If $t \le Y$, then $A(t) = 0$. Let $t > Y$ and $Q = \lfloor t/R \rfloor \ge 1$. By Dirichlet's theorem there are coprime a, r with $1 \le r \le Q$ and $|\beta - a/r| \le 1/(r(Q+1))$; then $|\beta - a/r|\, t \le R$. If $r \le R$, then Lemma 7.3 with $b = a$ gives $|A(t)| \le C_2 k t$, where

$$k = \frac{R \log(qR+2)(2+2\pi R)}{Y^{1/20} \log Y}.$$

If $r > R$, then $R \le r \le t/R$ and $|\beta - a/r| \le r^{-2}$, so Theorem 6.1 for $f = \chi$ bounds the full sum $\sum_{n\le t} \chi(n) e(n\beta)$, and since $\log t \le L$, removing the $\Psi(t) = \#\{n \le t : P(n) \le Y\}$ smooth terms gives

$$|A(t)| \le C_1\Big(\frac{t}{\log t} + t\rho\Big) + \Psi(t), \qquad \rho = \frac{L^2}{R^{1/4}} = \frac{L^2}{M^4}.$$

Since $tw(t) \le 1/t$ and $w(t) \le 1/t^2$,

$$\sum_{1\le t<H} |A(t)|\, w(t) \le C_1 \sum_{2\le t\le H} \frac{1}{t\log t} + (C_1\rho + C_2 k)\sum_{t\le H}\frac{1}{t} + \sum_{1\le t<H}\Psi(t)w(t).$$

The first sum is at most $3+\log L$, the second at most $1+L$, and by partial summation the third is at most $\sum_{n\le H,\,P(n)\le Y} 1/n \le e^{11}\log Y$. Finally $L \le M \le 2L$ gives $\rho(1+L) \le 2/M \le 128$; $\log(qR+2) \le 18M$, $2+2\pi R \le 8M^{16}$, $Y^{1/20} = M^{35}$ and $\log Y = 700\log M \ge 700$ give $k(1+L) \le 288M^{34}/(700M^{35}) \le 1$; and $\log Y = 700\log M \le 1400\log L$. Collecting terms, the sum is at most $1 + 131C_1 + C_2 + (2800e^{11} + C_1)\log L \le (2800e^{11} + 1 + 132C_1 + C_2)\log L$. □

Proof of Theorem 1.7. The character $\bar\chi$ is primitive modulo q. If $\log q \ge 200$, then Proposition 7.4 for $\bar\chi$ and Lemma 7.1 give

$$\left|\sum_{n\le x}\chi(n)\right| \le \sqrt{q}\,(C_4\log\log q + 6) \le (C_4+6)\sqrt{q}\log\log q.$$

If $\log q < 200$, then every sum $\sum_{m\le H}\bar\chi(m)e(m\beta)/m$ with $H \le q$ has modulus at most $1+\log q < 201$, and Lemma 7.1 gives

$$\left|\sum_{n\le x}\chi(n)\right| \le 207\sqrt{q} \le \frac{207}{\log\log 3}\sqrt{q}\log\log q.$$ □

8. Formal verification and computations

The Lean 4 development (Lean v4.34.1, Mathlib v4.34.1) proves

```
theorem least_prime_primitive_root_bound {η : ℝ} (hη : 0 < η) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧
      (ℓ : ℝ) ≤ C * (((p - 1).divisors.card : ℝ) ^ 2 *
        Real.log (p + 2)) ^ (8 + η)
theorem least_prime_primitive_root_le_rpow {ε : ℝ} (hε : 0 < ε) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧ (ℓ : ℝ) ≤ C * (p : ℝ) ^ ε
theorem erdos_985_of_large : ∃ p₀ : ℕ, ∀ p : ℕ, p.Prime →
    p₀ ≤ p → ∃ q : ℕ, q.Prime ∧ q < p ∧
      orderOf (q : ZMod p) = p - 1
```

which are Theorem 1.1 and Corollaries 1.2 and 1.3, and

```
theorem lindelof {w : ℝ} (hw : 0 < w) :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ ≠ 1 → ∀ s : ℂ, 7 / 8 + w ≤ s.re →
        ‖DirichletCharacter.LFunction χ s‖ +
          ‖(DirichletCharacter.LFunction χ s)⁻¹‖ ≤
          C * ((q : ℝ) * (3 + |s.im|)) ^ w
theorem character_sum_bound {w : ℝ} (hw : 0 < w) :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ ≠ 1 → ∀ x : ℝ, 1 ≤ x →
```

```
        ‖∑ n ∈ Finset.Icc 1 ⌊x⌋₊, χ n‖ ≤
          C * (q : ℝ) ^ w * x ^ (7 / 8 + w)
theorem character_sum_le_rpow {η θ : ℝ} (hη : 0 < η) (hθ : 7 / 8 < θ) :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ ≠ 1 → ∀ x : ℝ, 1 ≤ x → (q : ℝ) ^ η ≤ x →
        ‖∑ n ∈ Finset.Icc 1 ⌊x⌋₊, χ n‖ ≤ C * x ^ θ
```

(namespace `Vinogradov.CharSum`), which are Theorems 1.4 and 1.5 and Corollary 1.6 (there $x \geq 1$ follows from $x \geq q^\eta$), and

```
theorem weak_mv : ∃ C : ℝ, 0 < C ∧ ∀ (f : ℕ → ℂ),
    (∀ m n, f (m * n) = f m * f n) → (∀ n, ‖f n‖ ≤ 1) →
    ∀ (α : ℝ) (a : ℚ) (N : ℕ) (R : ℝ), 2 ≤ R → R ≤ a.den →
      (a.den : ℝ) ≤ N / R → |α - a| ≤ 1 / (a.den : ℝ) ^ 2 →
        ‖∑ n ∈ Finset.Ioc 0 N, f n * e (n * α)‖ ≤
          C * (N / Real.log N + N * Real.log N ^ 2 / q4 R)
theorem polya_vinogradov_loglog :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ.IsPrimitive → 3 ≤ q → ∀ T : ℕ,
        ‖∑ n ∈ Finset.Ioc 0 T, (χ n : ℂ)‖ ≤
          C * Real.sqrt q * Real.log (Real.log q)
```

(namespaces `Vinogradov.MV` and `Vinogradov.PV`, with $e(x) = \exp(2\pi i x)$ and $\mathtt{q4}\, x = \sqrt{\sqrt{x}}$), which are Theorems 6.1 and 1.7; the reduced fraction a/r is the rational number `a` with denominator `a.den`. Theorem 1.8 enters as the theorem

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`

of [19], stated for Mathlib's `DirichletCharacter.LFunction` and compiled from its source. The Lean proofs of Lemmas 2.4 and 5.1 use the logarithmic control lemma of [19] in place of the Borel–Carathéodory and three circles arguments, and the proof of Lemma 3.3 uses Mathlib's bound $\psi(y) \geq (y-1)\log 2 - \log(y+2)$ in place of [18]. The Lean proof of Lemma 6.5 uses Mellendijk's formalisation of Selberg's sieve in [22], and that of Lemma 6.4 bounds the prime powers by $\sum_n 4n^{-3/2}$. Every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Programs in Python, C, C++ and Julia compute $g^*(p)$ for all primes $3 \leq p \leq 10^8$ (Remark 1.9), check Lemma 3.1 numerically for $p \leq 200$, the inequality of Lemma 3.3 on $10^4 \leq x \leq 2 \cdot 10^7$, Lemma 3.2 for $N \leq 10^7$, the exponent identity in the proof of Theorem 1.1 in exact arithmetic, the two values in Remark 1.11, the identity in the proof of Lemma 5.2 in exact arithmetic for six characters, the inequalities in the proof of Proposition 5.3 at $2 \cdot 10^5$ random points, the choices of parameters in Section 5 in exact arithmetic, $|\tau(\chi)|^2 = q$ and Lemma 7.1 for the quadratic characters modulo the primes $q \leq 3000$, the bound $|r(\theta)| < 3$ in its proof on a grid of 10^6 points, the inequalities between the parameters in the proof of Proposition 7.4, Lemmas 6.2 and 6.3 on random data, Lemma 6.4 for 60 completely multiplicative functions, Lemma 6.5(i) for $X \leq 10^5$ and even $0 < |h| \leq 100$,

Lemma 6.5(ii) for $X \le 10^5$, and the numerical constants and the choice of I and J in the proof of Theorem 6.1. For the quadratic characters modulo the primes $10^3 \le q \le 2 \cdot 10^4$ the ratio $\max_x |\sum_{n \le x} \chi(n)|/(\sqrt{q} \log\log q)$ of Theorem 1.7 is at most 0.848.

Data availability

The Lean project, the programs and a script that runs them are archived at Zenodo, doi:10.5281/zenodo.0000000.

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